Article Review

A Machine Learning System for Precipitation Estimation Using Satellite and Ground Radar Network Observations

Oscar Ramirez Moguel, Rafsan Al Amin, and Sepehr Fekri

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Introduction



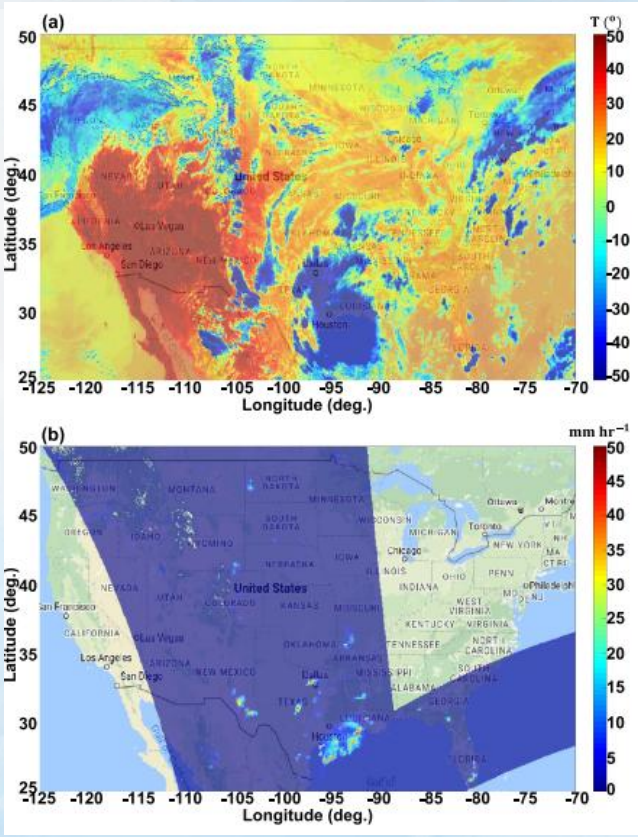
Introduction

Title: A Machine Learning System for Precipitation Estimation Using Satellite and Ground Radar Network Observations.

Authored by: Haonan Chen, V. Chandrasekar, Robert Cifelli, and Pingping Xie.

Year of Publication: 2019

- Proposes a machine-learning-based data fusion system
- Combines satellite observations with ground radar measurements
- Aims to improve satellite-based precipitation estimates



a) IR brightness temperatures b) PMW-based rainfall retrievals

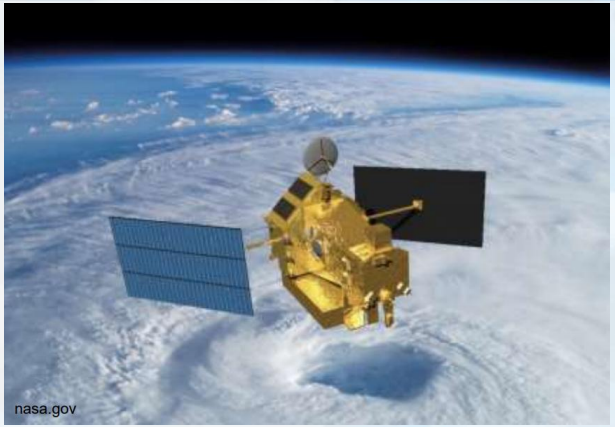
Introduction

Why is it important to improve satellite-based precipitation estimates?

- Precipitation plays a key role in understanding the water cycle
- It is a critical input for climate, hydrological, and weather forecast models

Methods of measuring precipitation

- Conventional meteorological stations (rain gauges)
- Meteorological radars
- Satellites: geostationary infrared (IR) and passive microwave (PMW) measurements







Introduction

Combination of weather radar data and satellite data for better precipitation estimates

Weather Radar

Advantages:

- High-resolution area coverage
- High temporal resolution
- Microphysics information (internal storm structure and raindrop size)

Drawbacks:

- Limited spatial coverage.

Satellite

Advantages:

- Near-global coverage
- IR-based algorithms - adequate for convective rainfall
- PMW-based measurements – more physically based / more accurate

Drawbacks:

- Spatio-temporal sampling gaps
- IR-based algorithms are not adequate for stratiform/warm-top rainfall
- PMW-based measurements have low temporal resolution

Introduction

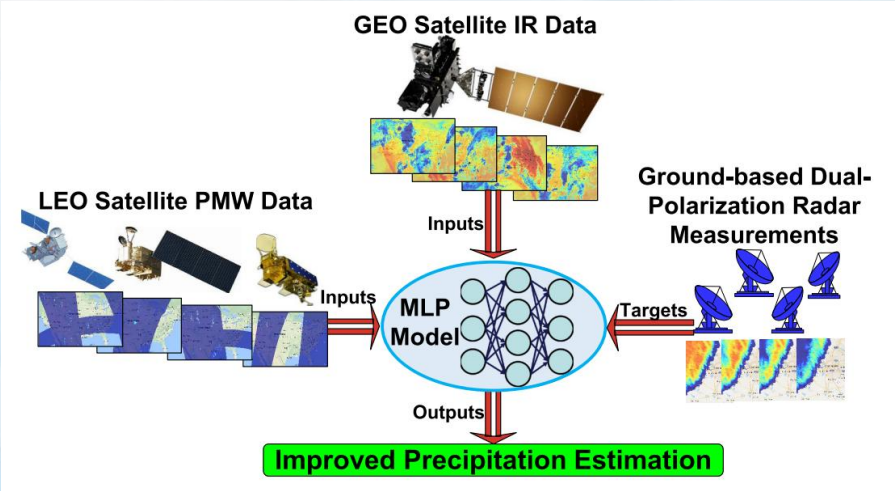
*IR = Infrared
*PMW = Passive Micro-Wave

Objective:

To introduce an innovative machine learning-based mechanism to improve spaceborne precipitation retrievals by incorporating dual-polarization measurements from a ground-based radar network

Key ideas:

- Designing a machine learning system for satellite-based precipitation estimation
- Using combined IR + PMW satellite data as inputs
- Using ground-based dual polarization radar measurements as training targets



02

Methodology



Methodology

Machine Learning Model

- **Multi-Layer Perceptron (MLP):** deep-learning model used to process the satellite data and produce precipitation estimates.
- **Architecture:** multiple layers of artificial neurons; input data (satellite IR and PMW) pass through several hidden layers before generating the output (precipitation estimates).
- **Training:** the MLP is trained in a supervised way, using high-quality ground radar rainfall estimates as target labels.

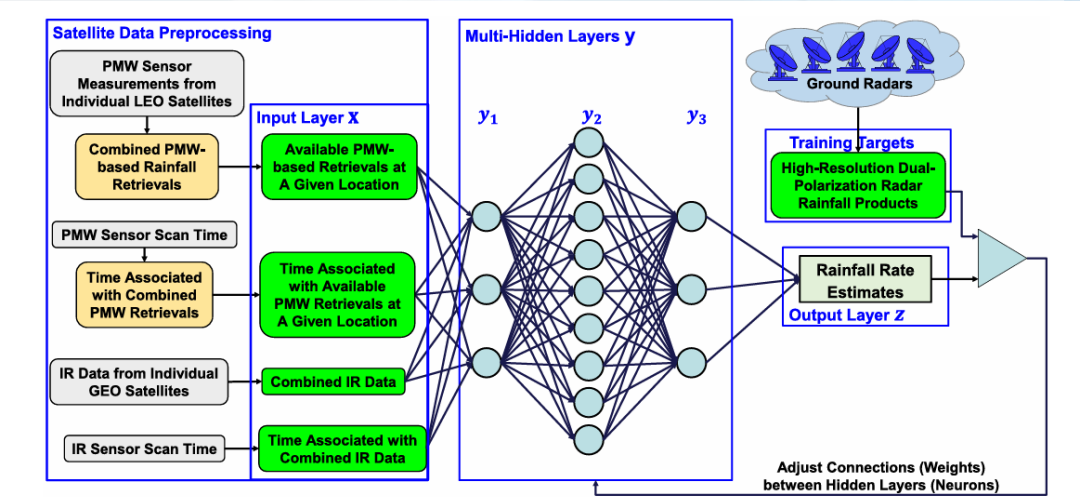


Fig. 2. Architecture of the deep learning model for satellite-based precipitation estimation using ground radar observations as references.

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$$\mathbf{y}_1 = f(\mathbf{w}_1\mathbf{x} + \mathbf{b}_1)$$
$$\mathbf{y}_2 = f(\mathbf{w}_2\mathbf{y}_1 + \mathbf{b}_2)$$
$$\mathbf{y}_3 = f(\mathbf{w}_3\mathbf{y}_2 + \mathbf{b}_3)$$
$$\mathbf{z} = f(\mathbf{w}_4\mathbf{y}_3 + \mathbf{b}_4)$$

Methodology

Inputs

1. Satellite Data

❑ **Infrared (IR) data**

from geostationary satellites (e.g., Himawari-8, GOES)

❑ **Passive Microwave (PMW)** data

from ***Low Earth Orbit (LEO)*** satellites (e.g., NOAA, MetOp satellites).

2. Radar Data

Ground Radar Data *(Target Labels)*

The machine learning model learns:
"Given this IR and PMW signal pattern, radar said rainfall was X mm/hr."

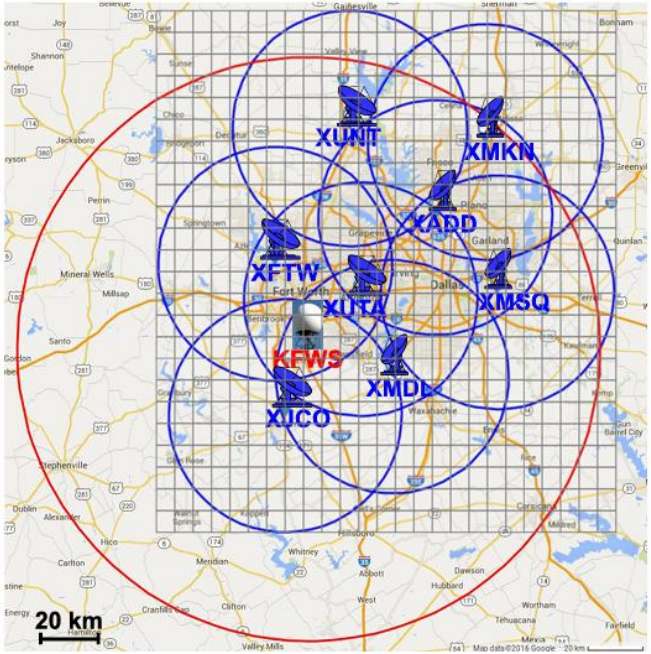
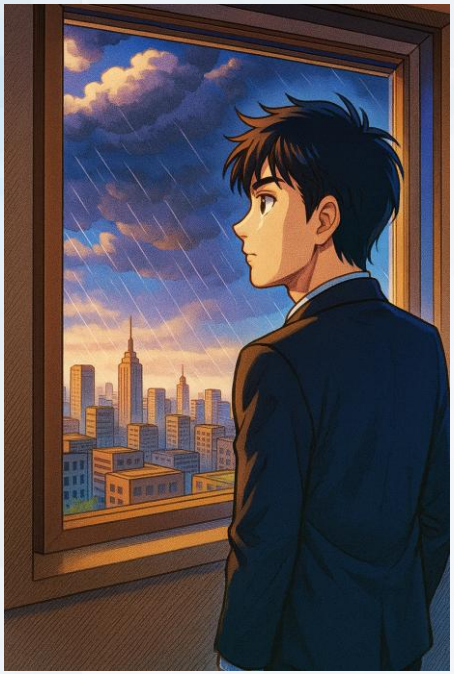


Fig. 4. Urban scale study domain for the proposed MLP-based precipitation estimation system. Circles in red (i.e., S-band) and blue (i.e., X-band) denote the layout of the DFW radar network. The letter symbols, such as "KFWS," correspond to the naming of various radars. The grids on top represent a 192 km × 200 km area, where this demonstration study will be focused on.

Dallas–Fort Worth Radar Network

Methodology







IR & PMW Satellite Observation

Truthing with Ground Radar Data

Methodology

IR vs PMW Comparison

Feature	IR	PMW
Frequency	Very frequent (every 30 min)	Rare (every 3–6 hours)
Accuracy	Low to medium	High
What it Detects?	Cloud-top temperature	Actual raindrops & ice inside the cloud

Methodology

Ground Radar Network Data

- High-resolution **dual-polarization radars**

(X-band + S-band, DFW network)

- Provides **near-surface, minute-by-minute rainfall estimates**
- Used as **ground-truth labels** to train the ML model
- Captures storm structure, raindrop size distribution, and intensity

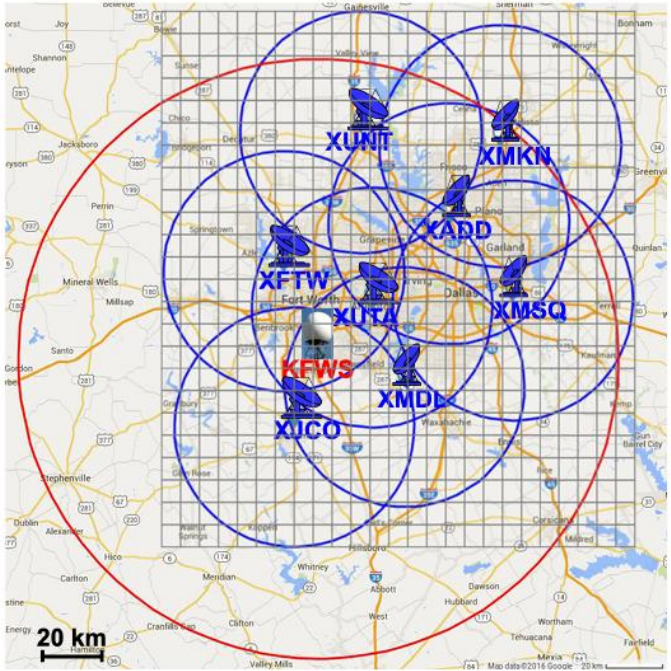


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Methodology

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Preprocessing and Data Fusion

Everything have to match PMW spatial resolution: 8 km × 8 km.

📌 **Radar (250 m → 8 km) (1 min → 30 min) (Upscaling)**

📌 **IR (4 km → 8 km) (30 min→ 30 min) (Upscaling)**

📌 **PMW (already 8 km) No change (3hour→ → → 30 min)**

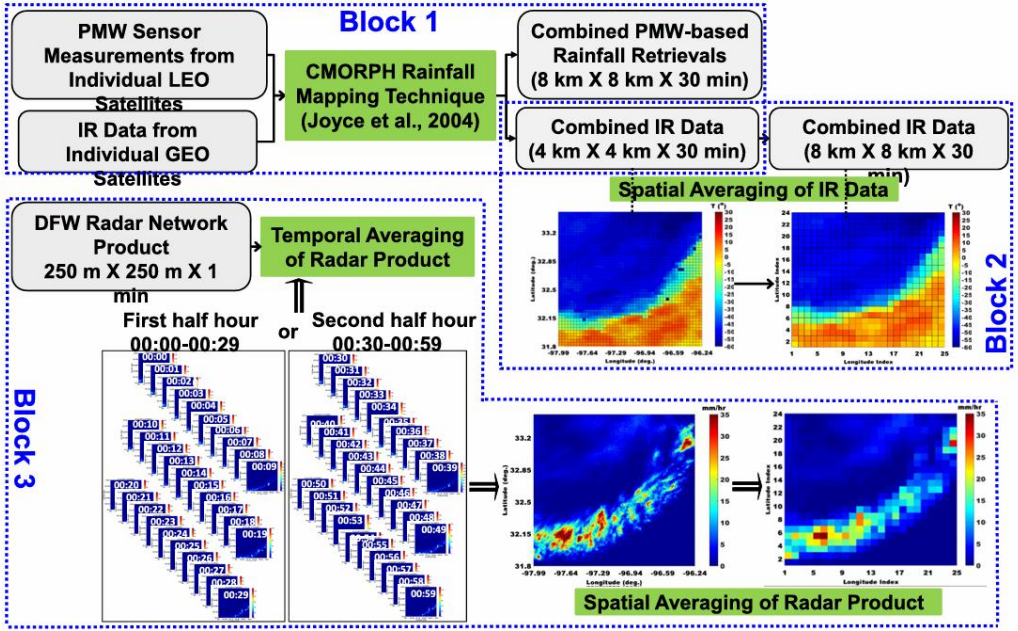


Fig. 7. Radar and satellite data preprocessing, including the spatial averaging of radar and satellite IR data, and temporal averaging of radar-based rainfall measurements. The preprocessed data will be ingested by the MLP model of the machine learning system in Fig. 2.

**Upscaling = making things more coarse (less detailed)*

Methodology

If PMW is not available at time T:

look at **PMW from −1.5 hours to +1.5 hours**

→ all available PMW data in that 3-hour window

Feed those values + the **time difference** as input to ML model

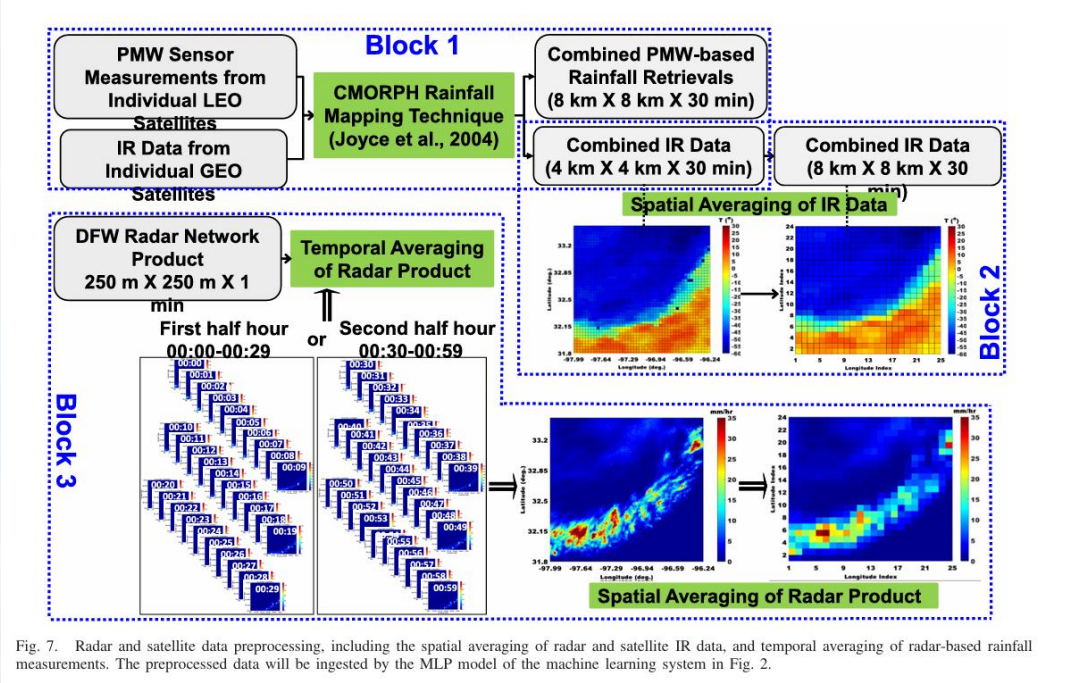


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Methodology

Training & Testing Events

Training Dataset

12 storm events from **2013–2015**

Covers a wide range of:

- Seasonal conditions
- Storm intensities
- Different rainfall patterns

Provides **~960 hours** of data for model learning.

Radar rainfall used as **ground truth** during training

Testing Dataset

2 completely unseen events

Used only to **evaluate** model performance.

Ensures the model can generalize to **new storms**.

Compared against:

- Radar rainfall
- CMORPH products

TABLE I

TRAINING AND TESTING EVENTS FOR THE PROPOSED DEEP LEARNING SYSTEM

Precipitation Dates	Category
17-20 Apr 2013	Training
15-18 May 2013	Training
26-29 Oct 2013	Training
16-20 Jul 2014	Training
17-20 Mar 2015	Training
01-04 Apr 2015	Training
08-15 Apr 2015	Training
10-13 May 2015	Training
28-31 May 2015	Training
16-20 Jun 2015	Training
25-29 Nov 2015	Training
25-29 Dec 2015	Training
08-11 Jun 2013	Testing
23-27 Jun 2014	Testing

Results

Left. Precipitation products on June 24, 2014, at 23:00 UTC
At this time frame, the combined PMW-based retrievals are available in the area

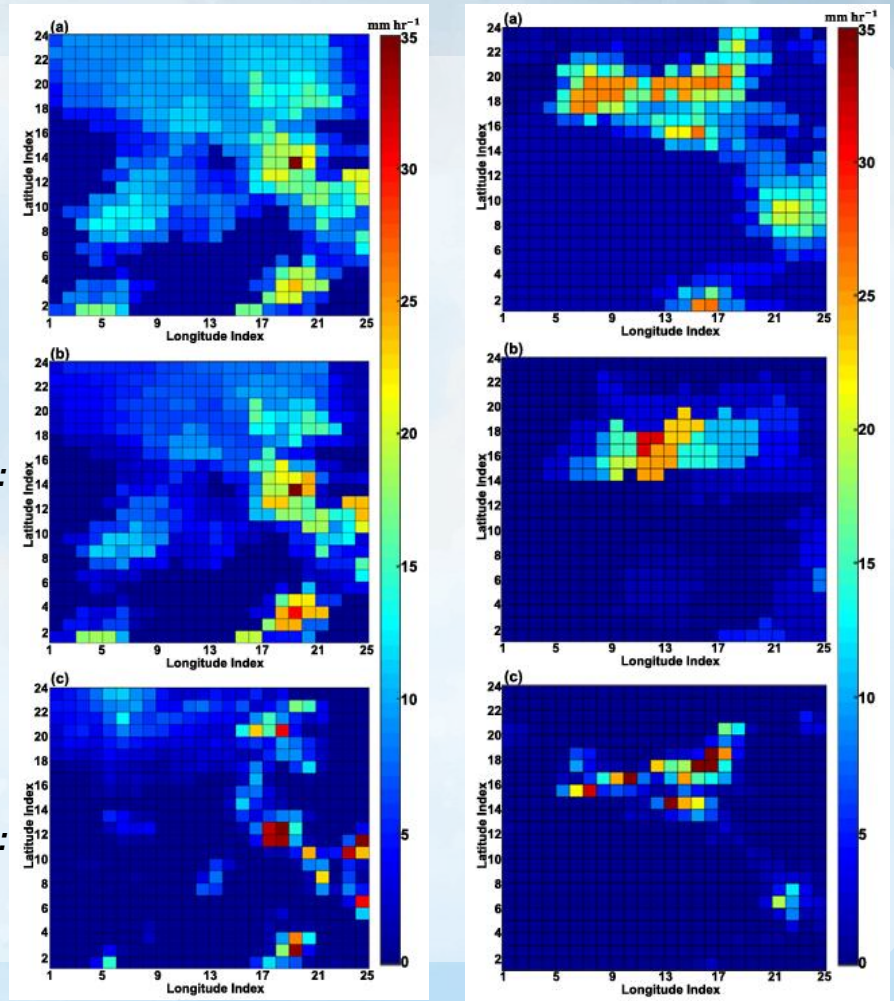
Right. Precipitation products on June 25, 2014 at 19:00 UTC
At this time frame, the combined PMW-based retrievals are not available in the area

- CMORPH and the MLP model could capture the overall rainfall pattern
- Time difference inputs allow the MLP to mimic CMORPH motion vectors

MLP:

CMORPH:

Ground Radar (truthing):



Results

$$\mathrm{NSE}_{\mathrm{rain}} = \frac{|E_R - R_R|}{R_R}$$

E_R = Number of rainy pixels from models

R_R = Number of rainy pixels from radars (ground truth)

Closer to zero = Better performance

MLP product outperforms the CMORPH product in two thresholds.

TABLE II

NSE OF RAINFALL ESTIMATES FROM THE PROPOSED DEEP LEARNING MLP MODEL AND THE STANDARD CMORPH SYSTEM. THE RESULTS SHOWN HERE ARE FOR THE TWO TESTING EVENTS LISTED IN TABLE I

	Normalized Standard Error (*NSE*)	
Threshold (mm hr^{-1})	CMORPH	Deep learning Model
0.5	15%	11%
1.0	40%	37%

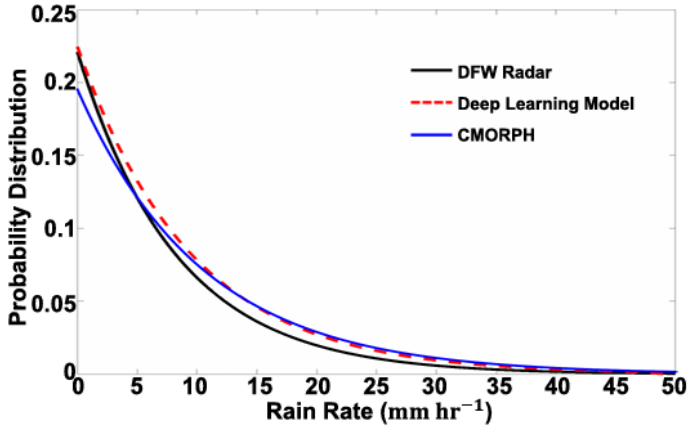


Fig. 11. Probability distribution of the estimated rain rates from the ground-based DFW radar network, the proposed deep learning model, and the standard CMORPH system for two testing events listed in Table I.

Conclusion & Observation

*DFW : Dallas–Fort Worth Radar Network

Main Findings:

- ❑ MLP model estimates precipitation from satellite data
- ❑ Performance is very similar to CMORPH product
- ❑ Time difference as input enables motion tracking

Limitations & Future Work:

- ❑ Trained only on local DFW observations → Need data from other regions for generalization
- ❑ Simple architecture (only 3 hidden layers) → Modern architectures (CNN, ConvLSTM)
- ❑ Basic evaluation metrics → Could use more comprehensive metrics
- ❑ Processing Power and Time Requirement

Strengths:

- ❑ Flexible - can be used wherever radar networks exist
- ❑ Well-documented procedure that can be followed
- ❑ Novel approach: time tags replace motion vectors



Thank you for your attention

